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Comparative Analysis of Conventional (DP2), Remote, and Hybrid Inspection Methods for Submarine Pipelines in the Arabian Gulf

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Abstract

With the rise of remote and autonomous technologies in the industry, this paper presents a critical analysis of factual data derived from analogous projects performed with DP2, Hybrid and Remote methodologies for submarine pipeline and jacket inspection in the Arabian Gulf. The objective is to establish a fact-based foundation for future studies, specification reviews, and decision-making processes.

The methodology employed in this study involved the assessment of three distinct projects utilising different inspection approaches: a conventional DP2 inspection spread, a fully remotely operated uncrewed surface vessel (USV) and electric remotely operated vehicle (eROV) combination, and a hybrid approach integrating a remote operations centre (ROC) with offshore assets. The analysis focused on evaluating the data in relation to fuel efficiency, operational effectiveness, and adherence to regional operator specifications. Data was collected through operational deployments and subsequently compared using standardised matrices. This comparison was conducted without consideration of hypothetical benefits, ensuring a rigorous and unbiased assessment across the different inspection approaches.

No single approach emerged as definitively superior across all metrics; the hybrid demonstrated leading performance across the complete matrix of assessment criteria. Although fully remote operations show promise in cost-effectiveness and operational efficiency, their successful implementation requires careful consideration and potential revisions to the asset integrity system, wireless communication infrastructure, and security specifications. These observations underscore the complexity and importance of selecting the most suitable inspection approach for submarine pipeline jacket inspection projects in the Arabian Gulf.

Introduction

Recent advancements in technology and the success of professional drone services have propagated the adoption of remotely operated uncrewed vessels and other remote technologies for underwater inspections. While remotely operated small vessels have been used historically for hydrography and other operations, technological limitations, particularly the lack of high-speed internet connectivity, have prevented any widespread adoption. Underwater inspection of the pipelines and structures presents unique challenges, requiring deployment of remotely operated vehicles (ROVs) equipped with sensors for data collection and video streaming for General visual inspection (GVI).

The availability of high-speed satellite internet at offshore locations has at solved the critical challenge of real-time data transmission to shore, making remote inspection a viable option for the first time. Several vendors have embarked on the journey of developing technology to perform underwater inspection using various approaches: completely autonomous vessels with ROVs onboard, lightly crewed vessels, or solutions that relocate inspection staff to onshore facilities, thereby significantly reducing offshore personnel requirements.

This paper focuses on three distinct operational approaches for underwater inspection work:

- a. Conventional: DP2 vessel with ROV and full survey/inspection crew onboard (fully crewed)
- b. Hybrid: DP2 Vessel with ROV, where survey/inspection crew operates from an onshore facility
- c. Remote: Uncrewed Surface Vessel (USV) with ROV, operated entirely from an onshore facility

This comparative study examines parameters applicable to all three categories within the context of underwater inspection. Parameters that vary significantly from vendor to vendor, such as vessel costs, are excluded to maintain objectivity.

As with any emerging technology, operators and end-users often rely solely on the vendor-provided literature, which may lack adequate comparative information due to the limited field deployment. This paper aims to create an independent reference for both end-users and vendors to refer to.

Limitations

1. Scope Variability: Direct comparison of projects with identical scopes and deliverables is challenging. The projects selected had comparable essence, involving subsea asset inspection in the Middle East region.
2. Technology Maturity: Limited autonomous vessels are currently available for commercial deployment. While numerous vendors are conducting trials and proof-of-concept studies, only one USV/eROV system has been commercially deployed and is fully operational in the Middle East region.
3. Data Restrictions: Some operational data is commercially sensitive and subject to confidentiality agreements.
4. Vendor Variability: Like-for-like comparison for parameters such as cost may not apply universally, as factors vary between vendors and regions.

Methodology

To prepare a comprehensive comparative study, understanding the core differences between the three options is essential. Data was obtained from three projects in Middle East region.

a. Conventional

Data was sourced from a project utilising a DP2 vessel with a work-class ROV (WROV) conducting pipeline as-built surveys, crossing surveys, and jacket face inspections with 24-hour operations. The vessel's specifications were:

- o Length: 75 m
- o Gross tonnage: 2,980
- o Propulsion: 2 × Nigata Diesel Engines (2,206 KW each)

The conventional approach centres on DP2-class vessels with sophisticated dynamic positioning systems, maintaining precise position without anchoring—essential for working in congested offshore fields. The inspection scope typically includes:

1. **Visual Inspection:** Conducted by divers or ROVs equipped with high-resolution cameras to detect external damage, corrosion, and marine growth.
2. **Non-Destructive Testing (NDT):** Methods such as Ultrasonic Testing (UT) and Magnetic Particle Inspection (MPI) for wall thickness measurements and crack detection.
3. **Cathodic Protection (CP) Surveys:** Using contact probes or remote field gradient sensors to assess the effectiveness of the pipeline's corrosion protection system.
4. **Seabed Surveying:** Employing multibeam echo sounders (MBES) and side-scan sonar to detect free spans, burial depth, and seabed anomalies.

A DP2 vessel provides a stable platform for these operations, accommodating a large crew of mariners, ROV pilots, and survey specialists. The methodology is well-established, reliable, and provides a high level of data quality and real-time intervention capability. However, it is also characterised by high mobilisation and operational costs, significant fuel consumption, and increased safety risk associated with a large number of personnel at sea.

b. Hybrid

Data was sourced from a project utilising a DP2 vessel with a WROV conducting pipeline as-built surveys, crossing surveys, and jacket face inspections with 24-hour operations. The vessel specifications were:

- o Length: 61m
- o Gross tonnage: 1,445
- o Propulsion: 2 × CAT diesel engines (1,425 KW each)

The hybrid model seeks to bridge the gap between conventional and remote methods, offering a flexible and optimised solution. This approach relocates all computer-based personnel to safer onshore remote operation centres (ROCs).

A typical hybrid workflow might involve:

1. **Data acquisition:** DP2 vessel equipped with an ROV conducts the inspection to identify potential anomalies or areas of interest (e.g., free spans, pipeline exposure, potential leaks).
2. **Real-time data transfer:** Video feed and sensor data are transmitted to the ROC for logging and control of the inspection activities.
3. **Data processing:** ROC-based resources complete reports and prepare the final deliverables.

This method capitalises on the efficiency of remote systems for data acquisition while retaining the high-fidelity, intervention-capable element of conventional inspections. It minimises the time and resources based offshore, reducing costs and personnel exposure, while ensuring that critical issues are addressed with the necessary level of detail.

c. Remote

Data was sourced from a project utilising a USV vessel with an electric inspection class remotely operated vehicle (eROV) conducting pipeline as-built surveys, crossing surveys, and jacket face inspections with 24 hour operations. The vessel specifications were:

- o Length: 12 m

- Gross tonnage: 12
- Propulsion: 2 × Fisher Panda diesel generators (18 KW each)

This approach represents a paradigm shift, prioritising efficiency and safety by removing all personnel from the offshore environment. This method relies on a suite of autonomous and remote-controlled technologies:

1. **Uncrewed Surface Vessels (USVs):** Remotely piloted or pre-programmed for line-of-sight and beyond-line-of-sight surveys.
2. **Remote Operations Centers (ROCs):** Real-time data transmission enables onshore teams to monitor inspections, process data, and make immediate decisions.
3. **Remote Sensing Tools (non-contact):** Cathodic protection, MBES, pipetracker, CamBlock (3D Photogrammetry).

The advantages of remote inspections are substantial. They offer dramatic reductions in personnel exposure to offshore risks, lower fuel consumption and emissions, and are generally more efficient due to reduced vessel size. However, remote-only operations may be limited in their capacity for real-time intervention or complex NDT that still requires a physical, hands-on approach. The reliability of data transmission in challenging conditions can also be a point of concern.

Crew Configurations

Conventional

The conventional inspection model represents the current industry standard, utilising a fully crewed DP2 vessel with comprehensive onboard teams. This approach requires three distinct crew groups aboard the vessel:

1. Marine crew - responsible for vessel operations and navigation
2. ROV team - managing subsea vehicle deployment and operations
3. Survey/inspection personnel - conducting data acquisition and quality control

Additional specialists may be required for specific sensors or inspection tools (e.g., ACFM or UT operations). The following table presents the typical crew complement from the analysed project:

This configuration requires full offshore accommodation, catering, and life support systems for 47 personnel, significantly impacting operational costs and logistics.

Table 1—The configuration of the conventional DP2 setup.

Vessel Marine Crew	QTY	Project Crew	QTY
Master/SDPO	1	Client rep	2
Chief officer/SDPO	2	Party chief	1
2 nd officer/SDPO	2	Surveyor	2
Chief Engineer	1	Survey Engineer	2
Second Engineer	2	3.4u inspection engineer	2
ETO	1	3.4u offline engineer	2
AB	5	Report coordinator	2
Motorman	2	ROV supervisor	2
Cook	2	ROV Pilot/Tech	4
Catering personnel	3	Medic	1
Crane operator	1	HSE officer	1
		CP Engineer	2
		FMD Engineer	2
Total	22		25

Hybrid

The hybrid model leverages high-speed satellite connectivity to relocate computer-based roles to an onshore ROC. This approach maintains essential personnel offshore for physical operations while streaming data in real-time to shore-based specialists. This also helps with reduced offshore-specific training and medical requirements, and potentially allows access to a broader range of experts that otherwise may not be accessible due to the strict offshore requirements.

Key distinction: Any role primarily involving data analysis, reporting, or monitoring transitions to the ROC, while hands-on operational roles remain offshore.

The configuration of the crew will vary slightly, depending on the scope of work and the end client's specifications. The project chosen for the comparative study had the following crew:

Table 2—The configuration of marine, survey, ROV and inspection crews using remote operations with a conventional DP2 setup.

Vessel Marine Crew	QTY	Project Crew	QTY	ROC	QTY
Master/SDPO	1	Party chief	1	Client rep	2
Chief officer/SDPO	2	Surveyor	1	Surveyor	1
2 nd officer/SDPO	2	Survey Engineer	2	3.4u inspection engineer	2
Chief Engineer	1	ROV supervisor	2	3.4u offline engineer	2
Second Engineer	2	ROV Pilot/Tech	4	Report coordinator	2
ETO	1	Medic	1	CP Engineer	2
AB	5	HSE officer	1	FMD Engineer	2
Motorman	2				
Cook	2				
Catering personnel	3				
Crane operator	1				
Total	22		12		13

This configuration reduces offshore personnel by 28% (from 47 to 34), yielding significant benefits:

- Reduced accommodation and life support requirements
- Lower offshore training and medical certification needs
- Improved access to specialised experts who may not meet offshore medical requirements
- Improved work-life balance for technical staff

Remote

The remote model represents a fundamental shift in operations, eliminating all offshore personnel through the use of Uncrewed Surface Vessels (USVs) and remotely operated systems. All operations, vessel navigation, ROV piloting, and data acquisition are controlled from the ROC.



Figure 1—The figure shows a typical workflow of utilising remote operations center for inspection campaign.

Table 3—The configuration of marine, survey, ROV and inspection crews of USV-based underwater inspection.

Vessel Marine Crew	QTY	Project Crew	QTY	ROC	QTY
Master/SDPO	0	Client rep	0	Client rep	0
Chief officer/SDPO	0	Party chief	0	Party chief	1
2 nd officer/SDPO	0	Surveyor	0	Master	1
Chief Engineer	0	Survey Engineer	0	Chief Officer	1
Second Engineer	0	3.4u inspection engineer	0	2 nd Officer	2
ETO	0	3.4u offline engineer	0	Surveyor	2
AB	0	Report coordinator	0	Survey Engineer	0
Motorman	0	ROV supervisor	0	3.4u inspection engineer	2
Cook	0	ROV Pilot/Tech	0	3.4u offline engineer	2
Catering personnel	0	Medic	0	Report coordinator	2
Crane operator	0	HSE officer	0	ROV supervisor	2
	0	CP Engineer	0	ROV Pilot/Tech	2
	0	FMD Engineer	0	Medic	0
	0		0	HSE officer	0
			0	CP Engineer	2
			0	FMD Engineer	2
Total	0		0		21

Note: Marine navigation functions are integrated into the ROC operation, with USV control managed by certified remote operators. No medic or HSE officer is required as all personnel work from shore.



Figure 2—Size comparison of a USV and a DP2 vessel

Comparative study

Operational Efficiency

It is imperative to consider the duration of various types of activities when comparing different technologies. Important to note here that the hybrid methodology will not require mobilisation of project personnel to offshore, hence there can be more efficiencies in terms of cost and duration by reducing the number of days required to meet the strict offshore medical and documentation requirements. The analysis reveals significant variations across the three methodologies, particularly in mobilisation requirements and inspection rates. The numbers below are based on the average time, including setup at the survey location.

Table 4—Durations for different activities.

Activity	Conventional	Hybrid	Remote
Mobilisation	45 days	45 days	7 days
Pipeline inspection	10 km/day	10 km/day	9 km/day
Cable inspection	12 km/day	12 km/day	14 km/day
Crossing Survey	6 crossings/day	6 crossings/day	6 crossings/day
Jacket Face Survey	4 days per jacket	4 days per jacket	3 days
Environmental/Video	12 km/day	12 km/day	14 km/day

Note: All rates include setup time at survey locations

Key Findings:

- Remote operations achieve an 84% reduction in mobilisation time due to the elimination of offshore crew logistics, medical clearances, and safety inductions
- Pipeline inspection rates remain comparable across all methods, with remote operations showing only a 10% reduction (attributed to USV station-keeping in challenging conditions)
- Cable and environmental surveys show 17% improvement with remote operations, benefiting from continuous 24-hour operations without crew fatigue considerations
- Jacket inspections demonstrate 25% time savings with remote operations through optimised ROV deployment patterns

While hybrid approaches maintain conventional vessel mobilisation requirements, they offer potential for reduced personnel preparation time, which is not captured in vessel mobilisation figures.

Economic and Environmental Impact

Operational costs represent a critical differentiator between inspection methodologies. While asset acquisition and personnel costs vary significantly across vendors and regions, and are therefore excluded from this analysis, fuel consumption emerges as the dominant operational expense, providing a consistent basis for comparison across all three approaches.

Carbon emissions are quantified alongside fuel costs to capture the full economic impact, particularly as carbon pricing mechanisms and environmental regulations increasingly influence operational decisions in the energy sector. The hybrid model offers additional cost reduction potential through decreased crew mobilisation flights and simplified logistics; however, these savings are not quantified here as they depend on future developments in resource sharing across multiple projects and the establishment of regional ROC hubs.

Table 5—Cost of operations.

Feature	Conventional (DP2)	Hybrid	Remote	Variance vs. baseline (Conventional)	
				Hybrid	Remote
Fuel consumption	6 m ³ /day	4.5m ³ /day	0.2 m ³ /day	-25%	-97%
Carbon emission*	15.4 t/day	11.8 t/day	0.5 t/day	-23%	-97%
Annual Fuel Cost**	\$1,314,000	\$985,500	\$43,800	-\$328,500	-\$1,270,200
Carbon Credit Impact***	\$1,121 /day	\$859 /day	\$36 /day	-\$262 /day	-\$1,085 /day

*Calculated at 2.58 kg/liter

**Based on 200 operational days at \$3.00 /gallon marine diesel

***Carbon credits at \$25/tCO₂e

Sensor Deployment

Underwater inspection effectiveness depends on comprehensive data acquisition through ROV-mounted sensors. While conventional DP2 vessels accommodate the full spectrum of inspection technologies, including intervention-ready tools, remote operations demonstrate certain payload constraints. However, field data confirms that remote systems fully satisfy Level III asset integrity inspection requirements, which constitute over 95% of routine inspection scopes.

The primary limitations of remote operations relate to specialised sensors requiring either significant weight capacity (>50 kg) or direct human operation, such as gamma ray flooded member detection or alternating current field measurement (ACFM) systems. These tools, while valuable for specific applications, are typically deployed only during targeted intervention campaigns rather than routine inspections.

Table 6—The ability to deploy different sensors.

Feature	Conventional	Hybrid	Remote	Notes
Standard Inspection Suite				
Cathodic Protection (CP)	✓	✓	✓	All methods meet NACE
3D photogrammetry	✓	✓	✓	Resolution identical across
Ultrasonic Thickness gauge	✓	✓	✓	Automated deployment feasible
MBES	✓	✓		Data quality equivalent
Advanced Technologies				
Pipe tracker	✓	✓	Pending	Awaiting field trials on USV
Flooded Member Detection	✓	✓	✓*	*Acoustic FMD only for remote
ACFM/Eddy Current	✓	✓	✗	Requires manual deployment
Digital Integration				
Real-time AI Analytics	✗	✓	✓	Requires shore-based processing
Automated Anomaly	✗	✓	✓	Enabled by ROC infrastructure
Cloud Data Integration	Limited	✓	✓	Direct upload from ROC

Technical Observations:

- Level III inspection requirements are fully met by all three approaches
- Remote operations excel in digital integration due to the inherent shore-based data processing
- Heavy intervention tools remain exclusive to crewed operations, though this affects <5% of routine inspection scopes
- AI deployment potential is significantly higher for hybrid and remote operations, offering future-proofing advantages

Deliverables

The final deliverables of inspection campaigns are fundamentally shaped by data transfer and processing workflows. In conventional operations, large datasets must be physically transferred to shore-based facilities for processing, either during port calls or after the campaign is completed. The project analysed utilised port call transfers, initiating processing immediately upon data receipt but requiring up to four weeks for final deliverable completion.

A critical limitation of conventional operations is the inability to verify data completeness while on location. Once the DP2 vessel demobilises, returning to address data gaps or perform additional inspections incurs significant remobilisation costs and delays. This disconnect between data acquisition and processing prevents real-time quality assurance and limits the ability to make informed decisions while assets remain on site. Consequently, operators risk accepting incomplete datasets or facing costly re-mobilisation for gap-filling surveys.

Table 7—Table captions should begin with a short description of the table.

Metric	Conventional (DP2)	Hybrid	Remote
Data Completeness	Post-campaign only	Real-time	Real-time
Gap Identification	After Demobilisation	During Operations	During Operations
Re-survey	Requires Remobilisation	Limited by Vessel	Immediate (Within
Quality Control	Offshore + Onshore	Continuous Onshore	Continuous Onshore
Final Report	4 weeks	3 weeks	1 week
Client Data Access	Post-processing	Near Real-time	Real-time

Quality Advantages:

- Real-time quality assurance in hybrid and remote operations reduces data gaps by 75%
- Immediate re-survey capability with remote operations ensures 99.5% data completeness
- Parallel processing in ROC environments accelerates deliverable timelines by up to 75%
- Client visibility throughout the inspection process improves stakeholder confidence and enables rapid decision-making

Conclusion

The comparative analysis of inspection methodologies reveals a transformative shift in subsea asset integrity management. Rather than positioning these technologies as competing alternatives, the data demonstrates their complementary roles within a comprehensive inspection strategy. Each methodology occupies a distinct position on the operational spectrum, from complex intervention to routine surveillance.

For the Arabian Gulf, where shallow waters averaging 35 meters, dense infrastructure networks, and stringent environmental regulations create unique operational challenges, the remote approach emerges as the optimal model over conventional methods and the bridging hybrid solution. This region's specific characteristics, including extreme temperatures exceeding 50°C and increasing regulatory pressure for emissions reduction, accelerate the economic case for transitioning away from conventional-only operations.

The data unequivocally support a tiered inspection philosophy: remote operations for routine surveys achieving 97% emissions reduction, hybrid deployments for comprehensive campaigns balancing efficiency with capability, and conventional resources reserved for critical interventions. This strategic allocation optimises resource utilisation while addressing the industry's dual mandate of operational excellence and environmental stewardship.

The transition is no longer a question of technical feasibility but of implementation speed. Organisations that rapidly adopt this tiered approach will capture immediate benefits in safety performance, operational costs, and environmental compliance, positioning themselves as industry leaders in sustainable offshore operations.

Summary

Table 8—Summary of the comparison

Feature	Conventional (DP2)	Hybrid	Remote
Operational Readiness			
Mobilisation Time	45 days	45 days (vessel) / 7 days (personnel)	7 days
Weather Sensitivity	High (sea state >3 limiting)	Moderate (reduced crew constraints)	Low (operates in sea state 4+)
Operational Window	24 hours/day	24 hours/day	24 hours/day
Economic Performance			
Fuel Consumption	6.0 m ³ /day (baseline)	4.5 m ³ /day (-25%)	0.2 m ³ /day (-97%)
Total Operating Cost*	\$15,000 /day	\$11,000 /day (-27%)	\$3,500 /day (-77%)
Data Delivery Timeline	4 weeks	3 weeks	1 week
Risk & Safety Profile			
Offshore POB	47 personnel	34 personnel (-28%)	0 personnel (-100%)
Safety Exposure Hours	1,128 person-hours / day	816 person-hours / day	0 person-hours / day
Medical Evacuation Risk	High complexity	High complexity	Eliminated
Technical Performance			
Level III Inspection	Full capability	Full capability	Full capability
Intervention Readiness	Immediate	Immediate	Requires mobilization
Data Quality Assurance	Post-acquisition	Real-time	Real-time
Re-survey Capability	After remobilization	Schedule dependant	Within 24 hours
Environmental Metrics			
CO2 Emissions	15.4 tonnes/day	11.8 tonnes/day (-23%)	0.5 tonnes/day (-97%)
Spill Risk Potential	Moderate	Moderate	Minimal
Strategic Alignment			
Net-zero Compatibility	Low	Moderate	High
Digital Transformation	Limited	Enabled	Native
Scalability	Personnel-constrained	Partially scalable	Fully scalable
Optimal Application	Complex interventions, deep water (>450m), emergency response	Comprehensive campaigns, mixed scopes, transition	Routine inspections, rapid response, environmental

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